Keshav Memorial College Of Engineering
A unit of Keshav Memorial Technical Educational Society (KMTES)
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Face Recognition-Based Smart Attendance System

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Introduction

Problem Statement
Manual attendance is **inefficient**, **error-prone**, and susceptible to **proxy attendance**.

Why Important?
Automating attendance boosts **accuracy**, saves **time**, and prevents **malpractice**.

Objectives

- Automate attendance marking using **facial recognition**.
- Eliminate **proxy attendance**.
- Store attendance records securely and accessibly in **Excel** or a **database**.

Literature Survey

Lead Author: Dr. V. Suresh

Affiliation: Assistant Professor, Department of Information Technology, Anil Neerukonda Institute of Technology and Sciences, Visakhapatnam, India

Summary: Dr. Suresh developed a facial recognition attendance system leveraging Python and OpenCV. The system automates attendance by capturing images, detecting faces, and comparing them with a pre-stored database, thereby enhancing efficiency and reducing manual errors.

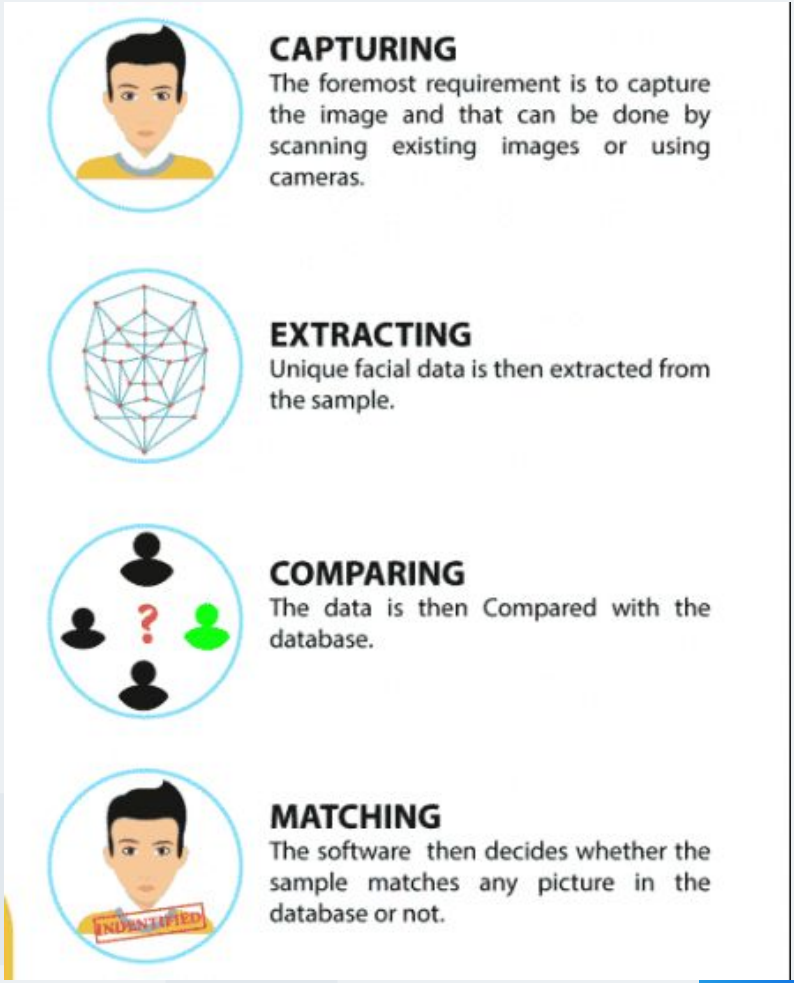
Lead Author: Dr. T. Chandrasekhar Rao

Affiliation: Professor, Department of Electronics and Communication Engineering, SV College of Engineering (SVCE), Tirupati, Andhra Pradesh, India

Summary: Dr. Rao's research introduces an automated attendance system utilizing facial recognition technology. The system employs image processing and machine learning techniques to accurately identify individuals and record their attendance, aiming to streamline the attendance process in educational institutions

Proposed Solution

- Utilizes **real-time camera feed** to detect and recognize faces.
- Compares with a trained **face dataset**.
- Marks attendance automatically if a **match** is found.
- Stores data in **Excel** for easy access.



System Architecture

Input Module

Camera captures real-time images.

Face Detection Module

Detects faces using Haar Cascade or DNN models.

Face Recognition Module

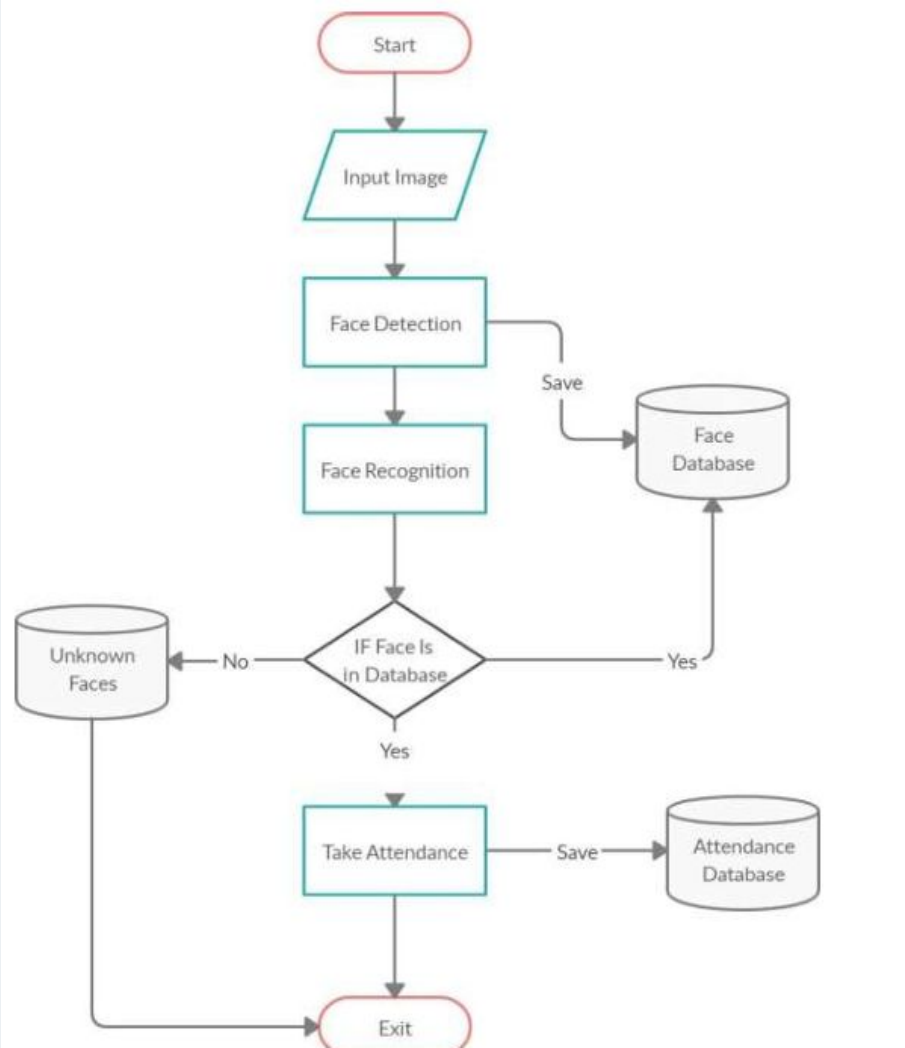
Extracts features via FaceNet or LBPH.
Compares with the **Face Database**.

Decision Logic

Match found → **Mark attendance**.
No match → **Log as Unknown Face**.

Data Storage

Face Database – Stores known faces.
Attendance Database – Excel/CSV/SQL with timestamps..



Pandas

Data analysis and manipulation

NumPy

Mathematical functions

matplotlib

SQLModel

Interact with SQL databases

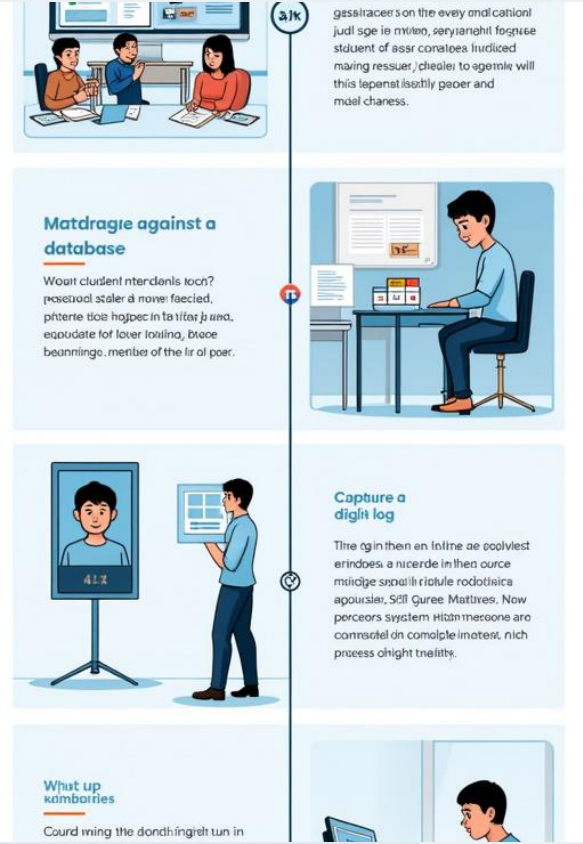
Modules & Tools Used

- **Programming Language:** Python
- **Libraries & Frameworks:**
  ->`OpenCv` – Real-time image processing & face detection
  ->`NumPy`, `Pandas` – Data handling & manipulation
  `Face Recognition` (Dlib-based or CNN-based model) – Facial feature encoding & matching
- **Deep Learning:**
  Pre-trained deep learning model for accurate facial recognition (ResNet or Dlib CNN)
- **User Interface:** Flask (Web-based UI for attendance view)
- **Data Storage:** CSV / Excel (for initial version), extendable to SQL/Cloud

Implementation / Working

- **Step-by-Step Process:**

1. **Student Registration**
   - Capture
   - and store face images in the system database.
2. **Attendance Capture**
   - Use live camera feed to detect faces in real time.
3. **Face Matching**
   - Recognize faces by comparing them with the trained dataset.
4. **Attendance Marking**
   - Mark presence and record data automatically in an Excel sheet.



Results

Performance Metrics:

- **Accuracy:** Achieved approximately **90% accuracy** for frontal facial images under controlled lighting conditions.
- **Speed:** The system performs **real-time face matching** with response times of **under 2 seconds**, ensuring minimal delay in marking attendance
- **Interface:** A user-friendly **recognition screen** displays live video input, detects faces, and confirms matches instantly with visual indicators.

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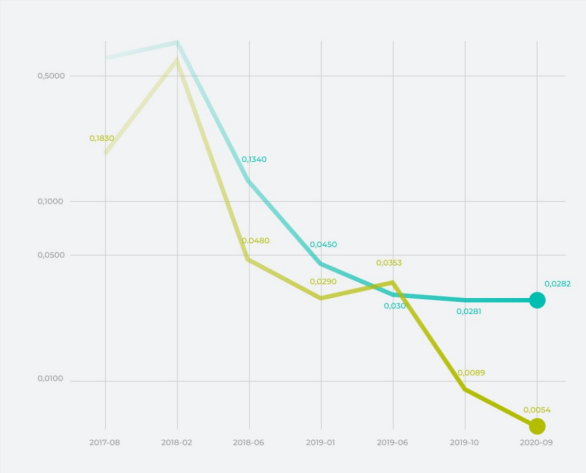

Challenges & Resolutions

Challenges Encountered

The main challenges encountered include **poor lighting conditions**, **head rotation during recognition**, and **CPU limitations** affecting performance.

Proposed Solutions

To address these challenges, the proposed solutions are to **preprocess images** for better quality, **implement optimal lighting setups**, and **utilize lightweight face models** for efficiency.

Future Scope

- **Enhance Security:** Integrate **liveness detection** (e.g., blink or motion verification) to prevent spoofing through photos or videos.
- **Improve Data Handling:** Transition from **Excel sheets to SQL databases or cloud storage** for scalable and secure data management.
- **Parental Notifications:** Implement **real-time alerts** via SMS or email to inform parents of student attendance status
- **Mobile App Integration:** Design a dedicated **mobile application for faculty**, enabling remote attendance access, monitoring, and reports

Conclusion

Our Face Recognition Based Attendance System delivers precise real-time attendance tracking with exceptional recognition accuracy and fast processing. Advanced image preprocessing and deep learning techniques effectively addressed challenges such as lighting variability, occlusions, and multi-face environments. This solution offers an efficient and user-friendly approach to automating attendance management across diverse settings.

Question & Answer

Thank you for your attention